

Discrete Mathematics
MATH1031 - Durham University

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Part I

Module Content

13 Introduction to Discrete Mathematics

13.1 Arrangements and Permutations

13.1.1 Arrangements

Definition 13.1.1.1: An arrangement is a sequence of items in which the order of the items in the sequence matters.

Definition 13.1.1.2: The length of an arrangement is the number of items in the sequence.

Standard Problem 1: How many arrangements of length k can be made from elements of a set S of size $|S| = n$?

Solution: n^k as arrangement has k places, each of which can be occupied by any one of n entries.

Example 13.1.1.1: How many four-letter “words” can be made from the standard A-Z alphabet?

Solution: $26^4 = 456,796$ words.

Standard Problem 2: How many arrangements of length k with no repeats can be made from elements of a set S of size $|S| = n$?

Solution:

$$n(n-1)(n-2)\dots(n-k+1) = \frac{n!}{(n-k)!}, \quad k \leq n \quad (1)$$

As we look at the available choices for each successive item in the list, the number of options is reduced by one each time.

Definition 13.1.1.3: The multiplication principle states that, for an arrangement of length k , if for $i \in \{1, 2, \dots, k\}$ there are n_i choices for the n^{th} item, regardless of previous choices, then the number of possible arrangements is $n_1 n_2 \dots n_k$.

13.1.2 Permutations

Definition 13.1.2.1: A permutation on a finite set S is an arrangement of length $|S|$ that uses each item from S exactly once.

Proposition 13.1.2.1: If $|S| = n$, then there are $n!$ permutations on S .

Proof: This is Standard Problem 2 with $n = k$.

□

Definition 13.1.2.2: We generalise this concept as, for a non-negative integer r , an r -permutation on S is an arrangement of r distinct elements from S . We write $P(n, r)$ for the number of r -permutations on a set of size n , which, by Standard Problem 2, we have,

$$P(n, r) = n(n-1)\dots(n-r+1) = \frac{n!}{(n-r)!}, \quad 0 \leq r \leq n \quad (2)$$

13.1.3 Counting Examples

Definition 13.1.3.1: The addition principle states that, for two disjoint, finite sets A and B . Then,

$$A \cap B = \emptyset \quad |A \cup B| = |A| + |B| \quad (3)$$

More generally, if $A_1, A_2, \dots, A_n$ are pairwise disjoint, then

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |A_1| + |A_2| + \dots + |A_n| \quad (4)$$

13.2 Combinations

Definition 13.2.0.1: A combination from a set of objects is an unordered selection of objects of that set.

Standard Problem 3: How many combinations without repetition do we have of a given size k from a set of size n ?

Solution: First consider that there are $P(n, k) = \frac{n!}{(n-k)!}$ arrangements without repetition. These are ordered lists, sorting them into groups of lists using the same letters will give groups of size $k!$. This means that the number of combinations is

$$\frac{P(n, k)}{k!} = \frac{n!}{k!(n-k)!} \quad (5)$$

Using the binomial coefficient notation, this means the number of combinations is $\binom{n}{k}$.

Example 13.2.0.1: How many ways are there of selecting ten different letters of the alphabet?

Solution: $\binom{26}{10} = \frac{26!}{10!16!}$.

Proposition 13.2.0.1: $\binom{n}{k} = \binom{n}{n-k}$

Proof: This can be seen from the formula for the binomial.

□

Proposition 13.2.0.2: Pascal's formula, for $1 \leq k \leq (n-1)$, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.

Proof: Proof excluded.

□

Theorem 13.2.0.1: The binomial theorem, if n is a positive integer and $a, b \in \mathbb{R}$, then

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} \quad (6)$$

Proof: The expression $(a+b)(a+b)\dots(a+b)$ when multiplied out, gives various terms of the form $a^k b^{n-k}$. For fixed k , the number of terms $a^k b^{n-k}$ is $\binom{n}{k}$ since one chooses the k brackets in the product form from which to choose the factor a rather than b .

□

13.3 Arrangements and Combinations with Repetitions

13.3.1 Arrangements with Repetitions

13.3.2 Combinations with Repetitions

13.3.3 The Extended Binomial Coefficients

14 Combinatorics

14.1 The Three Principles

14.2 Recurrence Relations

14.3 Generating Functions

15 Graph Theory

15.1 What is a Graph?

15.2 Eulerian Graphs

15.3 Planar Graphs

Part II

References

16 Module References

16.1 Module Professors

Introduction to Discrete Mathematics and *Combinatorics* are taught by Professor Mikhail Menshikov.

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Graph Theory is taught by Irene Ayuso-Ventura.

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16.2 Cited Works

n/a